Compile Autoware Algorithm

Disclaimer: This lesson has been successfully compiled on the Orin-Nano motherboard. Orin-Nano motherboard system environment: **Ubuntu22.04+ROS2-Humble**, the compiled autoware algorithm is **Autoware.Universe**. Since the actual motherboards used are different, the content of this chapter **only provides reference source code and reference tutorials**. It will explain how to install the environment and compile the source code according to the official steps. Any installation environment errors and compilation errors that occur need to be resolved by yourself.

Downloading Source Code

There are two ways to obtain the source code here. The first is to directly copy the [autoware] folder from the [autoware Source Code] folder in the [Attachments] folder. The second is to use git to pull the official source code. It is recommended to use the second method. If due to network reasons, you cannot pull the official source code, consider using the source code folder we provide.

Here is how to use git to pull the official source code. Enter the following command in the terminal,

```
git clone https://github.com/autowarefoundation/autoware.git
```

Installing Dependencies

Automatically install dependencies through the provided Ansible operation manual. Enter the following command in the terminal,

```
cd autoware
bash ansible/scripts/install-ansible.sh
ansible-galaxy collection install -f -r ansible-galaxy-requirements.yaml
#Choose 1 of the following 2 commands. If you support NVIDIA GPU installation, select 1. If you do not support NVIDIA GPU installation, select 2.
#1 allowable NVIDIA GPU
ansible-playbook autoware.dev_env.install_dev_env
#2 unallowable NVIDIA GPU
ansible-playbook autoware.dev_env.install_dev_env --skip-tags nvidia
```

Wait for the dependency installation to complete.

Setting Up the Workspace

Autoware uses [vcs2l] to build the workspace. Enter in the terminal,

```
cd autoware
mkdir -p src
vcs import src < repositories/autoware.repos
```

Install the dependent ROS packages. Enter in the terminal,

```
cd autoware
source /opt/ros/humble/setup.bash
sudo apt update && sudo apt upgrade
rosdep update
rosdep install -y --from-paths src --ignore-src --rosdistro $ROS_DISTRO
```

Compiling the Workspace

Autoware uses colcon to build the workspace. Enter in the terminal,

```
cd autoware
colcon build --symlink-install --cmake-args -DCMAKE_BUILD_TYPE=Release
```

Wait patiently for the compilation of 200+ packages to complete. Errors may occur during the process. If you want to skip the failed packages, add the compilation parameter --continue-on-error. The complete compilation command is as follows,

```
cd autoware
colcon build --symlink-install --cmake-args -DCMAKE_BUILD_TYPE=Release --continue-on-error
```

Reference Links

Autoware Installation: Installation - Autoware Documentation

Autoware.Universe Official Documentation: Common Documentation - Autoware Universe Documentation